6.3 Medians and Altitudes of Triangles

Name _____ Period ____ Date _____

Geometry 6.3 · Student sheet

I can…

- use the Centroid Theorem to find lengths on a median.
- find the centroid of a triangle in the coordinate plane.
- find the orthocenter of a triangle and tell where it is.

Vocabulary · fill in the missing word

Median of a triangle — a segment from a vertex to the _____ of the opposite side

Centroid — the point of _____ of the medians of a triangle

Altitude of a triangle — the _____ segment from a vertex to the opposite side or to the line that contains the opposite side

Orthocenter — the point where the lines that contain the _____ of a triangle intersect

Part 1 · Discovery: balance the triangle

Draw the three segments from each vertex to the midpoint of the opposite side. Measure in centimeters.

Segment from	Whole segment	Vertex to the meeting point	Meeting point to the midpoint	Ratio of the pieces
A				
B				
C				

Part 2 · Notes and examples (fill in each note, then work the example)

Centroid Theorem: the centroid is _____ of the distance from each vertex to the midpoint of the opposite side.

Example 1

Point P is the centroid of $\triangle LMN$, and $\overline{NQ}$ is a median. NQ = 18. Find NP and PQ.

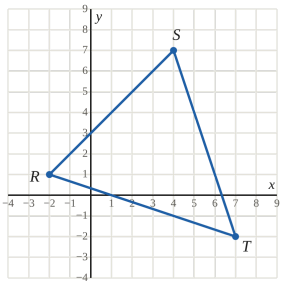
Example 2

Point G is the centroid of $\triangle ABC$, and $\overline{AE}$ is a median. AG = 3x − 2 and GE = x + 3. Find x and AE.

Centroid from coordinates: the centroid's coordinates are the _____ of the vertices' coordinates: $((x_1 + x_2 + x_3)/3, (y_1 + y_2 + y_3)/3)$.

Example 3

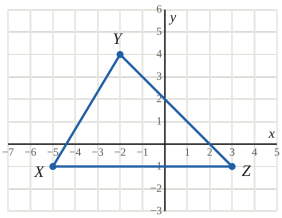
Find the centroid of the triangle with vertices R(−2, 1), S(4, 7), and T(7, −2).



Where the orthocenter is: inside an acute triangle, at the vertex of the _____ of a right triangle, and outside an obtuse triangle.

Example 4

Find the orthocenter of the triangle with vertices X(−5, −1), Y(−2, 4), and Z(3, −1). Is it inside, on, or outside the triangle?



Example 5

A triangle has vertices (0, 0), (8, 0), and (−2, 4). Its angle at (0, 0) is obtuse. Is each point inside, on, or outside the triangle?

a. the centroid

b. the orthocenter

Part 3 · Partner practice

Partner 1

Point P is the centroid of $\triangle LMN$, and $\overline{NQ}$ is a median. PQ = 5. Find NP and NQ.

Partner 2

Find the centroid of the triangle with vertices (1, 1), (5, 3), and (3, 8).